4TH ANNUAL ECE RESEARCH DAY

Quantifying Talker Familiarity Benefits in Cochlear Implant User Speech Perception – A Baseline Solution for Generative Voice Conversion

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1. Introduction

Motivation and Background

- The limited spectral resolution of the cochlear implant (CI) electrode array impairs speech perception, especially in background noise [1].
- **Talker familiarity advantage (TFA).** Familiar voices are easier to understand [2, 3].

Study Goal and Hypothesis

CI users understand voices they rate as familiar better, and generative voice conversion (VC) can carry part of that benefit to unfamiliar speech.

2. Methods

Listeners, Talkers and Tasks

Category	Description
Listeners	7 CI users, 10 normal-hearing (NH) controls
Talkers	12 public figures (6 male, 6 female) + 2 unknown
Tasks	Familiarity rating (1 to 10) + sentence transcription
Experiment 1	Natural speech, quiet and +10 dB SNR noise
Experiment 2	Sentences converted with StarGANv2-VC [4], quiet
Delivery	Remote MATLAB interface via AWS WorkSpaces

Three-Stage Pipeline

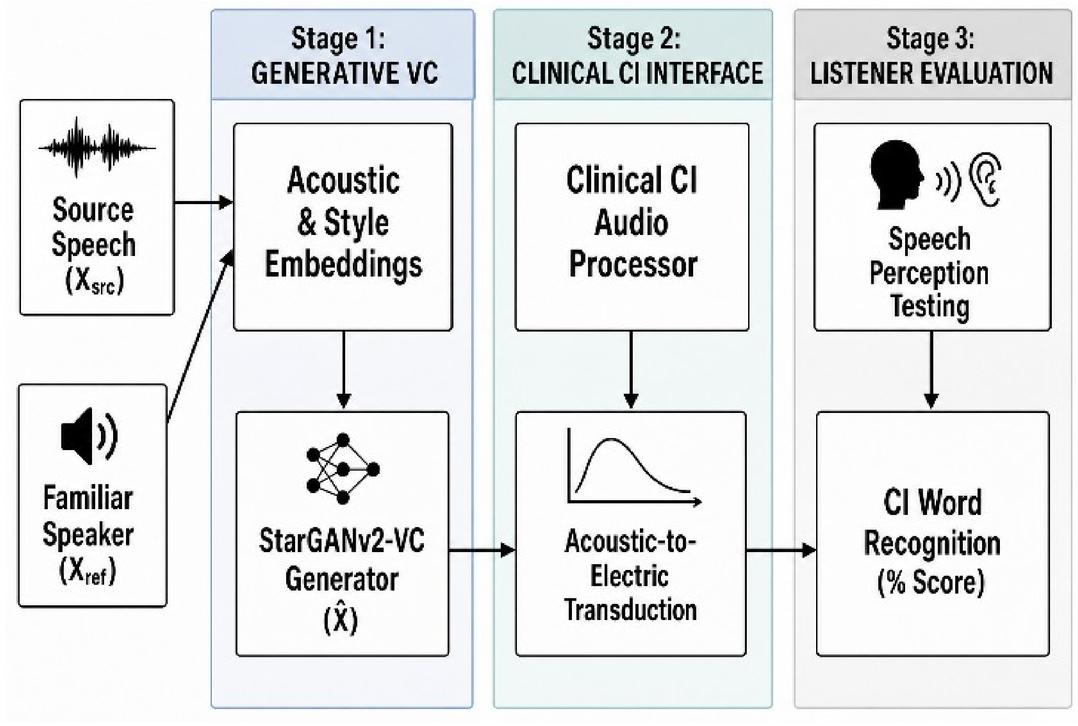


Fig. 1. From voice conversion to CI listener evaluation

3. Results – Familiar Voices

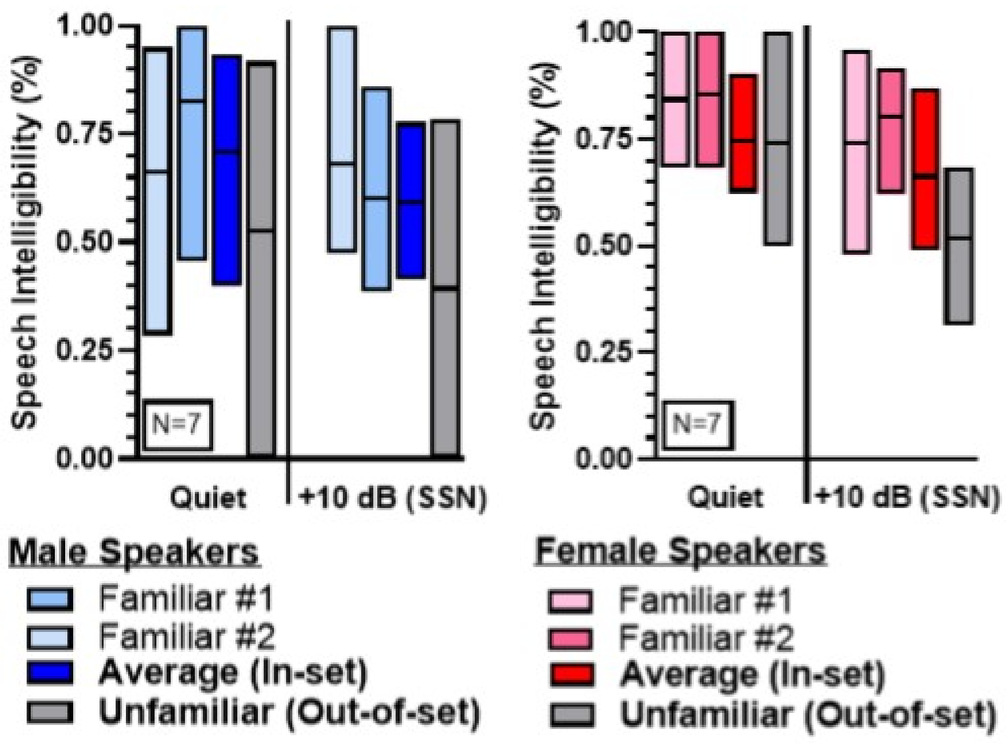


Fig. 2. Familiar vs. in-set average vs. unfamiliar talkers (N=7 CI)

Takeaway. **Familiar voices stayed near 80% in quiet, while the unfamiliar voice fell to about 55% (female) and below 50% (male) in noise.**

4. Results – Voice Conversion

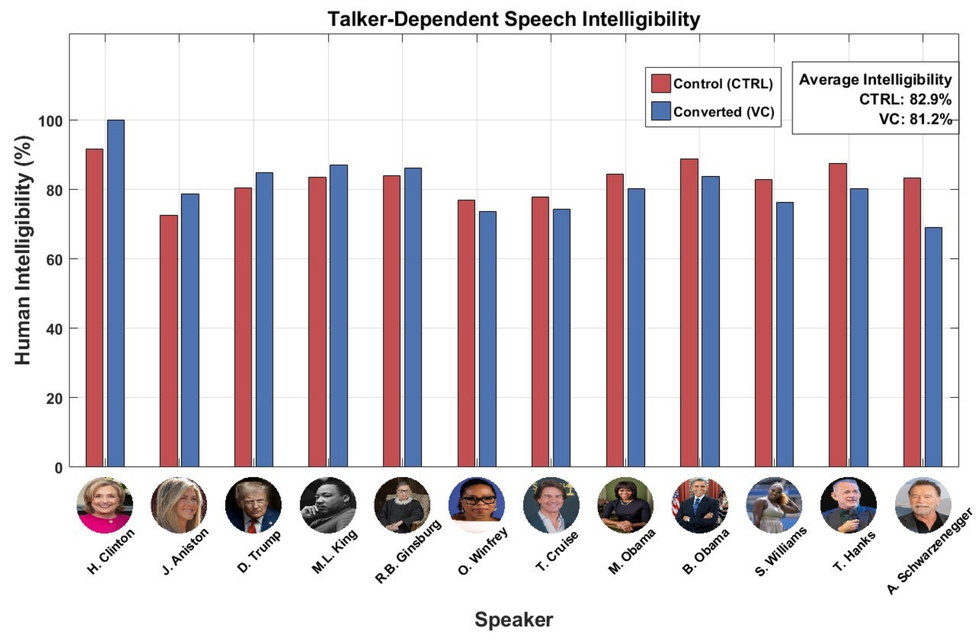


Fig. 3. CI word recognition, control vs. converted speech (N=7)

5 of 12
voices improved after conversion

+8%
best gain (Clinton)

−14.5%
largest drop (Schwarzenegger)

Takeaway. **The VC benefit is speaker dependent, so overall averages stay close (82.9% vs. 81.2%).**

5. Analysis – What Predicts a Gain?

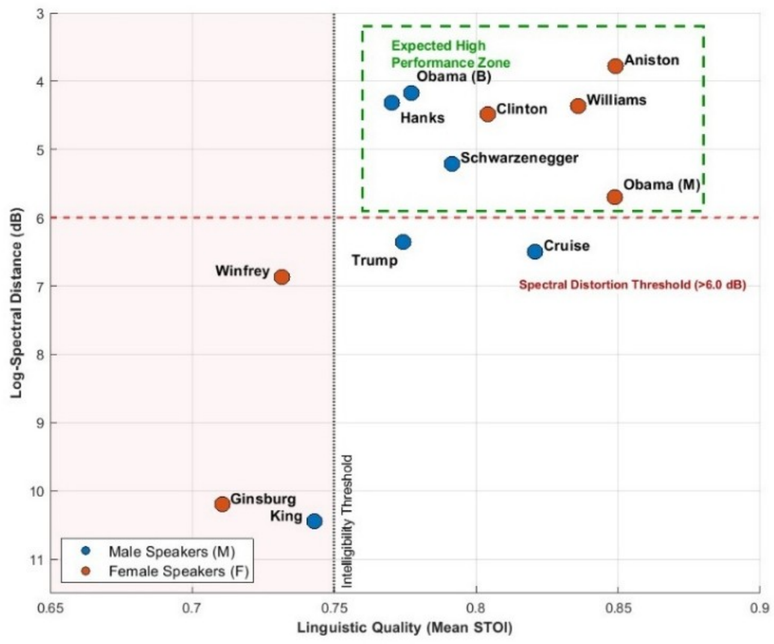


Fig. 4. Speaker acoustic landscape, STOI vs. LSD

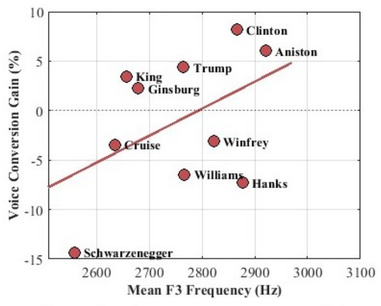


Fig. 5. Gain vs. mean F3

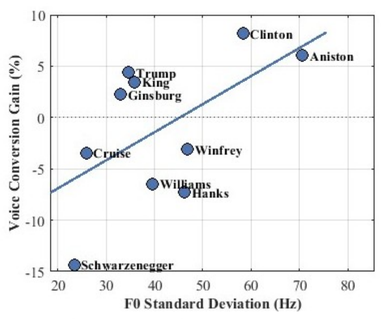


Fig. 6. Gain vs. F0 SD

- Narrow-pitch targets get compressed, blurring the envelope cues CI users rely on.

Takeaway. **High F3 (above 2800 Hz) and dynamic pitch (F0 SD above 55 Hz) predict a gain.**

6. Conclusions

- Familiar voices improve CI word recognition, most clearly in noise.
- VC transfers that benefit only for target voices that leave the model room to work.
- STOI and LSD alone do not predict how CI users hear converted speech.

Future Direction

- Add a CI-processing term to the VC training loss so converted speech keeps the slow temporal envelope.
- Build a one-knob familiar-voice dial that moves an unknown talker toward a voice the listener knows.

7. References

[1] D. Wang and J.H.L. Hansen, J. Acoust. Soc. Am., vol. 143, no. 4, 2018.
[2] P. Souza, N. Gehani, R. Wright and D. McCloy, J. Am. Acad. Audiol., vol. 24, no. 8, 2013.
[3] I.S. Johnsrude et al., Psychol. Sci., vol. 24, no. 10, 2013.
[4] Y.A. Li, A. Zare and N. Mesgarani, "StarGANv2-VC," 2021.

Forty Years
Electrical and Computer Engineering
1986-2026
2026